

CYFI725 2025 Midterm (100 Points)

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Instructions

- This is an open-book exam with a time limit. Each question is 20. No additional tools are required.
- Answer all questions using the **Linux Terminal** unless otherwise specified to use the Windows Command Prompt. Ensure you test your Linux VM before the exam.
- Discussions are prohibited during the exam. Do not share your results with others. Using generative AI to answer questions is **prohibited**.
- **MUST** only use the commands we covered in the class. **Justify** if you believe you have to use different commands to answer questions.
- You **MUST** verify if each command has been executed correctly by capturing screenshots of the execution results. **Highlighting** the correct answers. You must achieve correct results to receive credit. Justify if you believe there are no answers or investigation results.
- The Canvas Dropbox will close automatically at the deadline. Submit your work a few minutes early, as the Canvas server's time may differ.
- Instructions related to screenshots can be found here https://github.com/frankwxu/digital-forensics-lab-p2/blob/main/practice/Assignment_Instructions.pdf

1 File Operations

- a. Create a file (e.g., `myfile.txt`) with the string "Hello world!", your full name, and today's date as the file content. For example, if I create a file, the content should be "Hello world! Frank Xu. 11/16/2025"
- b. Change the file name to `file0Ps_yourlastname.txt`. Replace `yourlastname` with your real last name. For example, my file name is `file0Ps_xu.txt`

2 Understand Disk (USB) Image

- a. Download the image file `vusb.dd`: <https://raw.githubusercontent.com/frankwxu/digital-forensics-lab-p2/main/practice/vusb.dd> and verify the MD5: 8d171ad7397c8a3436073514269a7411

- b. What is the file system the USB uses?
- c. How many sectors does the USB have?

3 File Recovery

- a. List all deleted files. Explain why you think these files are being deleted.
- b. Recover files related to the password using TSK commands from `vusb.dd` and save the recovered file as **yourfirstname_recovered.txt**
- c. Are you able to recover the deleted file only using the sector numbers? Recover the file if your answer is "Yes", otherwise, explain why not?
- d. What is the timestamp of the most recent access to the deleted password file?

4 Understand How Information was Saved in Disk

- a. The recovered password file contains credentials for Sarah Smith. Report the exact physical location on disk where the password data was recovered, expressed as sector number and byte offset within that sector.

5 Basic Python

- a. What are the differences between Unix, Webkit, and Mac OS Epoch time?
- b. Calculate the difference between Unix and Webkit time in seconds. The output should be "The difference between two different epoch times is xxx seconds." You need to replace "xxx" with calculated results